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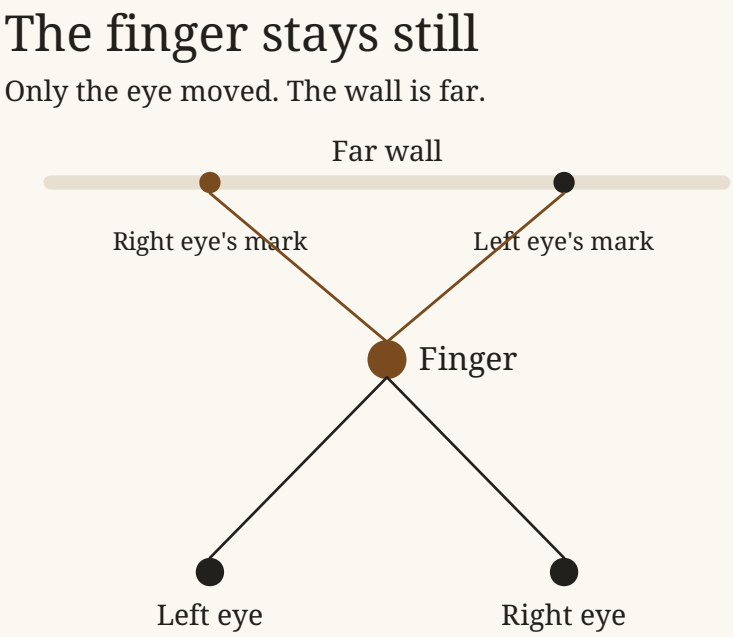
Hipparchus: the Moon changes place when we move

The first part of this page is a surveying idea, with a made-up steeple and made-up numbers. The second part is what the historical record actually gives for Hipparchus. Those are different jobs. The steeple is not his observation.

The shift you see is not the thing moving

Hold a finger at arm's length. Look with the left eye, then the right, and keep the head still. The finger jumps against the far wall. The finger did not move. You changed eyes. The wall is so far that it does not jump. The finger is near, so it does.



The finger stays still. Two eyes line one finger up with two different places on a far wall. The gap between the eyes is the baseline.

The gap between your eyes is the baseline. A baseline is a distance you already know, because you can measure it on yourself. The jump is a change in direction. Direction here means which way you are looking, lined up with the finger and a mark on the wall.

The Moon can do the same thing, if you move far enough. The Moon has not suddenly flown sideways. You moved. A far background, the Sun or the stars, stays put. The Moon, being nearer, shifts against that background.

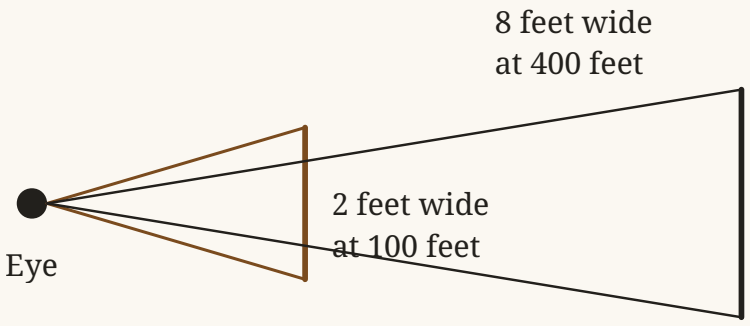
One look cannot tell size from distance

A degree is one three-hundred-sixtieth of a full turn. A full turn is the circle you would make by turning all the way around.

A sign 2 feet wide, 100 feet away, can fill the same angle as a sign 8 feet wide, 400 feet away. Four times the width, four times the distance, the same look. One glance cannot tell which sign it is. A single photograph of that look cannot tell either. The photograph records the angle. It does not record the miles.

Same look, four times as big

Four times the width at four times the distance.



One look cannot tell these apart.

A photograph of the width has the same limit.

This is a teaching picture, not a measurement of the Moon.

Same look, four times as big. A sign 2 feet wide at 100 feet and a sign 8 feet wide at 400 feet fill the same angle. One look cannot tell them apart.

A picture of the Earth's shadow on the Moon has the same limit. It can show that the shadow is about two and a half Moon-widths across. That is a ratio of widths, at whatever distance the Moon happens to be. It is not yet a distance in miles, and it is not yet the Moon's width in miles. A larger Moon, farther away, could fill the picture in the same way.

Two directions and a measured baseline

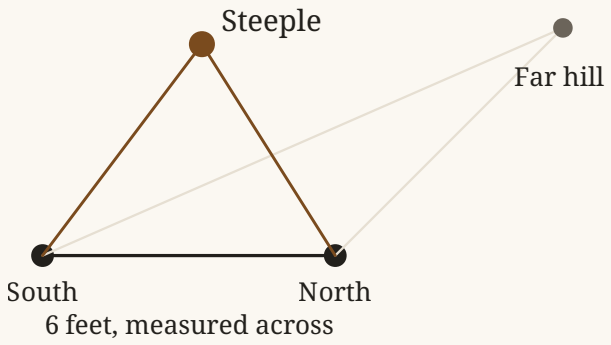
What breaks the ambiguity is a second place to stand, a known distance from the first, and a second direction.

Here is a teaching example. It is not Hipparchus, and it is not the Moon. South and North stand side by side, 6 feet apart. That 6 feet was measured on the ground, straight across the view, not along the line toward the steeple. That crosswise distance is the baseline. Both can see a steeple, and both can see a hill much farther away. A ridge a mile off is already about fifteen times farther than the steeple will turn out to be, so a 6-foot step does not shift the hill. The hill is the far wall.

Each person lines the steeple up against the hill and says how far around the steeple sits from the hill's direction. In this made-up case, the two answers differ by 1 degree. That 1 degree is the angle at the steeple, between the two sightlines. On the drawing the angle is large, so you can see the lines cross. A real degree is much thinner. The arithmetic below treats the 6 feet as a small piece of the circle around the steeple, which is fair when the baseline is crosswise and the angle is small.

Two directions, one baseline

A made-up steeple. Not Hipparchus, and not the Moon.



The hill does not shift: it is too far. The steeple does.
The two rust lines differ by 1 degree at the steeple.
Here that angle is drawn large so the crossing can be seen.
About 57 baselines fit into that distance.
57 times 6 feet is 342 feet.

Two measured directions. South and North are 6 measured feet apart, side by side. Their sightlines to a steeple differ by 1 degree against a much farther hill. The distance is about 342 feet. The angle is drawn large.

Now the distance, in words. A full turn is 360 degrees. The rim of a circle is about 6.28 times the distance from the middle to the rim. So one degree of the rim is 360 divided by 6.28, which is about 57 times shorter than that distance. Flip it: if the baseline covers 1 degree at the steeple, the distance to the steeple is about 57 baselines. 57 times 6 feet is 342 feet. That is a steeple a short walk away, not a steeple hundreds of miles off.

The hill is not infinitely far. Move 6 feet and a ridge one mile away shifts by about 0.065 degree. Set beside a 1-degree difference, that leftover is about 6 percent. Calling the hill "the far wall" is a small approximation, not an exact shared direction. The Sun, or the stars, is far enough that the leftover shift is negligible for this kind of picture. The circle count, 360 divided by 6.28, is itself about 57, not a sharper instrument.

Check the jobs. Measured: the 6 feet across, and each person's direction from the hill to the steeple. Calculated: the 1 degree difference, then the 342 feet. Not assumed: the steeple's height, or how far it was before you started. The ground between the two people is the ruler.

After the distance, the width

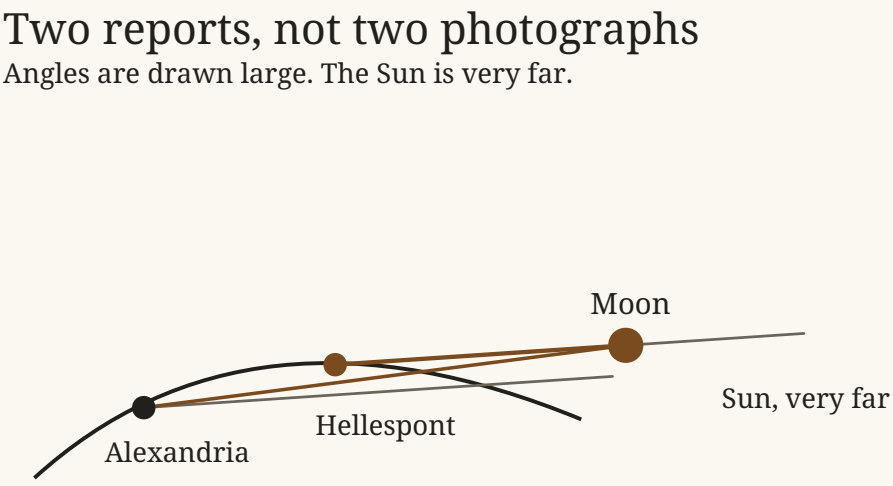
Once the distance is known, the look's width becomes a real width. Suppose the steeple also covers about 1 degree, the same amount as the shift between the two people. A whole degree at 342 feet is 342 divided by 57, about 6 feet. The steeple is about 6 feet wide. Measured: how wide it looks, in degrees. Already calculated: the 342 feet. Newly calculated: the 6 feet of real width. You did not need the width to get the distance, and you did not need the distance handed to you in advance.

That is the order. Baseline and two directions first. Distance second. Width in miles third.

What Hipparchus actually had

Hipparchus worked in the second century BCE. The book in which he treated the sizes and distances of the Sun and Moon is lost. What we have are later reports, chiefly Ptolemy in the Almagest and Pappus commenting on that passage. This is not a pair of star photographs taken at the same moment from two modern observatories. Nobody in that century had a camera. He used reports of how an eclipse looked.

Pappus reports one solar eclipse from Hipparchus's first book. Near the Hellespont the Sun's disk was entirely hidden. At Alexandria about four-fifths of it was hidden. The shift between those reports is about a fifth of the Sun's width in the sky. Read that as a fraction of the width, not of the area. Four-fifths of the area would be a different, larger bite.



Two reports, not two photographs. Alexandria and the Hellespont, the Moon, and a very far Sun. The angles are drawn too large. The difference in the eclipse is a difference of direction.

The Sun is the far wall. It is so far that the two cities can treat its direction as shared. The Moon is the finger. It is nearer, so the cities do not see it in the same direction against the Sun. The difference in the eclipse is that difference of direction.

The miles, or the Earth radii, enter with the baseline. An Earth radius is the distance from the Earth's center to its surface. The cities are a known kind of distance apart only because someone has measured the Earth and the cities' latitudes. Latitude is how far north or south a place sits. The place names alone are not a length. G. J. Toomer's reconstruction uses about 41 degrees for the Hellespont and about 31 degrees for Alexandria. Those are part of the modern reading of the geometry, not a sentence copied from a surviving page of Hipparchus.

In that first book, as Pappus reports it, the least distance of the Moon is 71 Earth radii and the greatest is 83. Pappus says the mean that follows is 77. Ptolemy says Hipparchus tried the problem by guessing at the Sun: in one try the Sun has no parallax you can perceive, which is the far-wall idea taken to the limit. Toomer's

1974 reading identifies the eclipse as 14 March 190 BCE and treats 71 as a minimum possible distance, because a simplifying assumption (the eclipse on the meridian at both cities) was not actually true. The year is Toomer's identification. The notice Pappus quotes does not itself print that date. This page will not pretend the year is secure.

A second book used a different guess about the Sun. Pappus reports the least lunar distance as 62 Earth radii, the mean as 67 and 1/3, and the Sun's distance as 490 Earth radii. The greatest lunar distance that belongs with those is 72 and 2/3. This is the shadow method. You assume how far the Sun is, which sets how the Earth's shadow tapers, and the Moon's distance follows. The assumed solar distance is doing the work that a measured baseline did in the other method. If the Sun is taken as infinitely far, Toomer notes that this shadow arithmetic levels off near 59 Earth radii. He reads Hipparchus as placing the mean between 67 and 1/3 and about 59. That "between" is Toomer's reading of the logic. It is not a sentence we have from Hipparchus in his own book.

Ptolemy, in the second century CE, later got a mean lunar distance of about 59 Earth radii from his own parallax measurement. That is his result in the Almagest. It is not a claim that he copied a number out of a lost page.

The same idea of a baseline shows up at other sizes. The Venus stations of 1761 and 1769 stand on the Earth's surface, a baseline of hundreds or thousands of miles. Kepler's Mars triangles use a baseline the size of Earth's orbit. Hipparchus uses the cities of two reports. A single triangle that only compares sizes does not by itself give a distance in miles.

The modern mean is about 60.3 Earth radii. NASA gives the Moon's average distance as 238,855 miles, which is 384,400 kilometers. The Earth's mean radius is 6,371 kilometers. 384,400 divided by 6,371 is about 60.3. The distance also changes through the month. Landing near that number is not the same thing as a tight measurement. Hipparchus's reported figures are 71 and 83 on one hypothesis, and 62, 67 and 1/3, and 72 and 2/3 on another. They are reconstructions from later writers, with an assumption about the Sun sitting inside the second set.

Sources

- Pappus, commentary on Almagest V, as quoted and translated in Christián C. Carman, "On the distances of the sun and moon according to Hipparchus," which prints the 71 and 83, the 62, the 67 and 1/3, and the solar distance 490.
- G. J. Toomer, "Hipparchus on the distances of the sun and moon," Archive for History of Exact Sciences 14 (1974): 126–142. The eclipse identification 14 March 190 BCE, the latitudes, and the reading of 71 as a minimum are his reconstruction.
- G. J. Toomer, "Hipparchus," Dictionary of Scientific Biography, the summary of that reconstruction, including the shadow limit near 59. MacTutor biography: Hipparchus.
- NASA Science, Moon facts, average distance 238,855 miles (384,400 kilometers).
- NASA Space Science Data Coordinated Archive, Earth volumetric mean radius 6,371.0 kilometers, on the Moon fact sheet.

Key facts

- A near thing shifts against a far background when the observer moves. The thing need not have moved.
- Four times the width at four times the distance can fill the same angle. One look, or one shadow picture, does not give miles.
- Two directions and a measured baseline fix the distance. The real width is calculated after that. The steeple numbers, 6 feet across and 1 degree and about 342 feet, are a teaching example. The 6 feet runs across the view.
- Hipparchus's book is lost. Pappus reports 71 and 83 Earth radii from a solar eclipse, total at the Hellespont and about four-fifths at Alexandria, and 62, 67 and 1/3, and a solar distance of 490 from a second hypothesis. The second uses an assumed solar distance.
- The year 190 BCE is Toomer's identification, not a date in the quoted notice. Ptolemy's own later mean is about 59 Earth radii. The modern mean is about 60.3.

The illustrated lesson is download the PDF.